

Hugo Flores García

email: hugggofloresgarcia@gmail.com

[check out my website!](#)

BIO

I perform research at the intersection of music, machine learning, and human computer interaction. I'm interested in building interfaces for musical expression, powered by deep learning.

EDUCATION

Northwestern University <i>Ph.D. in Computer Science</i>	Evanston, IL 2020 - 2025
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Georgia Southern University <i>B.S. in Electrical Engineering</i>	Statesboro, GA 2016 - 2020
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EXPERIENCE

Adobe Research <i>Research Scientist</i> Artist-centered controls for generative sound.	San Francisco, CA 2025.09 - Present
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Adobe Research <i>Research Intern</i> Advisors: Prem Seetharaman, Oriol Nieto, Justin Salamon. Sound generation via vocal imitations.	San Francisco, CA 2024.06 - 2025.02
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Descript <i>Research Intern</i> Advisor: Prem Seetharaman. Music Generation with Masked Transformers.	Remote 2022.09 - 2023.05
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Spotify <i>Research Intern, Audio Intelligence</i> Advisors: Rachel Bittner and Jan Van Balen. Text-guided audio effects.	Remote 2022.06 - 2022.09
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Northwestern University <i>Research Assistant, Interactive Audio Lab</i> Advisor: Bryan Pardo. Deep Learning Systems for AI Music Co-Creation.	Evanston, IL 2020.08 - present
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Audacity (Google Summer of Code) <i>Developer</i> Source Separation and Extensible Deep Learning Tools	Remote 2021.05-2021.09
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Georgia Southern University <i>Research Assistant</i> Advisor: Fernando Ríos. Accessible Digital Musical Instruments with EEG sensors.	Statesboro, GA 2018.08 - 2020.05
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SELECTED PUBLICATIONS

- H. F. García, O. Nieto, J. Salamon, B. Pardo, and P. Seetharaman. Sketch2sound: Controllable audio generation via time-varying signals and sonic imitations. In *ICASSP*, 2025
- H. Flores Garcia, P. Seetharaman, R. Kumar, and B. Pardo. Vampnet: Music generation via masked acoustic token modeling. In *ISMIR*, 2023
- D. Flores García, H. Flores García, and M. Riondato. Clavenet: Generating afro-cuban drum patterns through data augmentation. In *Proceedings of the 19th International Audio Mostly Conference: Explorations in Sonic Cultures*, AM '24, page 355–361, New York, NY, USA, 2024. Association for Computing Machinery

4. H. Flores Garcia, P. O'Reilly, A. Aguilar, C. Benetatos, Z. Duan, and B. Pardo. Harp: Bringing deep learning to the daw with hosted, asynchronous, remote processing. In *7th Workshop on Machine Learning for Creativity and Design at NeurIPS 2023*, 2023
5. Y. Wang, H. F. García, and J. Choi. *Few-Shot and Zero-Shot Learning for Music Information Retrieval*. In 23rd International Society of Music Information Retrieval Conference, 2022
6. H. Flores Garcia, A. Aguilar, E. Manilow, and B. Pardo. Leveraging hierarchical structures for few-shot musical instrument recognition. In *Proceedings of the 22nd International Society of Music Information Retrieval Conference (Best Paper Award)*, 2021
7. H. Flores Garcia, A. Aguilar, E. Manilow, D. Vedenko, and B. Pardo. Deep learning tools for audacity: Helping researchers expand the artist's toolkit. In *5th Workshop on Machine Learning for Creativity and Design at NeurIPS 2021*, 2021

ART INSTALLATIONS

Token Telephone

NIME 2024 Conference (Utrecht, NL), Experimental Sound Studio (Chicago, USA) 2024

Quadraphonic Interactive Neural Sound Installation. Collaboration with Stephan Moore. demo: <https://www.youtube.com/watch?v=vEaYoEgtSUo&t>

Salad Bowl

NeurIPS 2023 Creative AI 2023

Interactive Neural Sound Installation. Collaboration with Stephan Moore and Bryan Pardo.

SELECTED COMPOSITIONS

unsound objects

premiered at CLEAT series in Chicago, IL, USA March 2025

for text-prompted generative neural network and 8-channel digital mixer. published at ICMC 2025.

The Ritual - All in Good Time

exhibited at the Museum Folkwang in Essen, Germany Sept 2024

commissioned by Vietnamese media artist Ngoc Nau's for their 3-channel video installation All in Good Time (2024). OP-1, electronics, and RAVE models. Released with 1473 Records.

world of mouth

premiered at Experimental Sound Studio, Chicago Feb 2024

8 channel fixed media composition. Sonic environments built by vocal gestures processed by a generative model. Featured in UNPOP exhibition at Burning Man 2024 and UNPOP REDUX at the University of Alberta's Sound Studies Institute Gallery.

confluyo yo

premiered at ISMIR 2023 in Milan, Italy November 2023

for tenor saxophone and a generative sound model.

flowerbeds

premiered at Channel Noise 2019 at Georgia Southern University 2019

audiovisual live coding.

SELECTED PERFORMANCES

ICMC 2025

Boston, MA June 10th 2025

Improvisation with my generative text-to-sound instrument, unsound objects.

See <https://www.hugofloresgarcia.art/interfaces#unsound-objects>

Boston AI Music Meetup 2025

Boston, MA April 2025

Improvisation with my generative text-to-sound instrument, unsound objects.

CLEAT

Elastic Arts Chicago, Chicago

March 2025

Improvisation with my generative text-to-sound instrument, unsound objects.

Improvised Music Series

Elastic Arts Chicago, Chicago

November 2024

Improvisation with augmented/prepared guitar, electronics, and AudioStellar and RAVE models.

StretchMetal's Drone Rodeo

the Hideout, Chicago

March 2024

40 mins of improvised ambient with synths, electronics, neural networks (AudioStellar) and markov chains. full performance available at <https://www.youtube.com/watch?v=T6eGKgG7o0>.

Chicago Creative Machines

Experimental Sound Studio, Chicago

Feb 2024

improvisation with AudioStellar and bass guitar. full performance available at <https://www.youtube.com/live/Nfh1RH5k-bg?si=YeXQcjNr1NrtuqZ2&t=5001>.

ISMIR 2024

Politecnico Di Milano, Italy

Nov 2023

performed "confluyo yo" with Bryan Pardo, as part of ISMIR 2024's music program.

OPEN SOURCE SOFTWARE

unloop

Developer

2023 - 2024

Unloop is a looper pedal in Max/MSP that uses generative modeling to not repeat itself.

See <https://github.com/hugofloresgarcia/unloop.html>.

HARP

Technical Lead

2023 - 2024

HARP is a sample editor that allows for hosted, asynchronous, remote processing of audio with machine learning. See <https://github.com/audacitorch/HARP.html>.

nesquik

Nesquik is a vampnet-based audio effect that will transform any instrumental music audio into an "8-bit", NES-style chiptune. See <https://huggingface.co/spaces/huggof/nesquik>.

Audacity (Audio Editor)

Developer

2021 - 2022

Contributed a software framework that lets deep learning practitioners easily integrate their own PyTorch models into the open-source Audacity DAW. This lets ML audio researchers put tools in the hands of sound artists without doing DAW-specific development work.

See <https://interactiveaudiolab.github.io/project/audacity.html>.

torchopenl3

A PyTorch port of the OpenL3 audio embedding model.

Used as class materials for [CS 352 - Machine Perception of Music and Audio](#)

See <https://github.com/hugofloresgarcia/torchopenl3>.

Philharmonia Dataset

PyTorch dataset bindings for the Philharmonia Orchestra sound samples.

Used as class materials for [CS 352 - Machine Perception of Music and Audio](#)

See <https://github.com/hugofloresgarcia/philharmonia-dataset>.

TALKS

Expressive Sound Generative Modelling via Vocal Gestures and Sound Morphing <i>UC Berkeley CNMAT</i>	<i>Jan 29 2026</i>
Controllable and Expressive Generative Modelling for the Sound Arts <i>UCSD</i>	<i>April 8 2025</i>
Controllable and Expressive Generative Modelling for the Sound Arts <i>MIT CSAIL Invited HCI Seminar</i>	<i>April 8 2025</i>
Controllable and Expressive Generative Modelling for the Sound Arts <i>Boston AI Music Meetup</i>	<i>April 10 2025</i>
The Voice is the Interface and Other Techniques for VampNet <i>Bay Area Signal Hackers (BISH) BASH</i>	<i>August 1 2024</i>
Compositional Techniques for VampNet <i>AI Music Reading Group, MIT Media Lab</i>	<i>April 15 2024</i>
generative sound for the sonic arts! <i>Chicago Creative Machines, Experimental Sound Studio</i>	<i>Feb 25 2024</i>
writing about music is like dancing about architecture! <i>GLASS Human-Centered AI Music Symposium, Northwestern University</i>	<i>Jan 26 2024</i>
VampNet: Music Generation via Masked Transformers <i>Spotify MIQ Reading Group</i>	<i>September 6 2023</i>
Deep Learning for Music Interfaces <i>Universidad Nacional Autónoma de México (UNAM)</i>	<i>April 6 2022</i>
Leveraging Hierarchical Structures for Few-Shot Musical Instrument Recognition <i>ISMIR 2021</i>	<i>November 9 2021</i>
Deep Learning Tools For Audacity: Helping Researchers Expand the Artist's Toolkit <i>Bay Innovative Signal Hackers (BISH) Bash</i>	<i>October 27 2021</i>
Deep Learning Tools For Audacity: Helping Researchers Expand the Artist's Toolkit <i>Neural Audio Synthesis Hackathon (NASH) Workshop</i>	<i>December 12 2021</i>

HONORS AND AWARDS

ICASSP Outstanding Reviewer Award <i>ICASSP 2023</i>	<i>2023</i>
Best Paper Award - Leveraging Hierarchical Structures for Few Shot Musical Instrument Recognition <i>ISMIR 2021</i>	<i>2021</i>
Cognitive Science Fellowship <i>Northwestern University</i>	<i>2020 - 2021</i>
Lewis and Charlene Stewart Jazz Scholarship <i>Georgia Southern University</i>	<i>2016 - 2020</i>
Coastal Jazz Scholarship <i>Coastal Jazz Association</i>	<i>2019</i>

Undergraduate Research Grant*Georgia Southern University*

2018

Honors Program 1906 Scholarship*Georgia Southern University*

2016-2020

SKILLS

- **Programming Languages** - *Expert*: Python, C/C++, *Intermediate*: Javascript, Lua (norns)
- **Machine Learning** - *Expert*: PyTorch, libtorch, Scipy, Numpy, Scikit-learn
- **Creative Coding** - *Expert*: SuperCollider, Max/MSP/Jitter, *Intermediate*: OpenFrameworks, P5js, Pure-Data, JUCE
- **Music Production** - Logic Pro, Ableton Live
- **Languages** - I can read/write/speak English and Spanish natively.

TEACHING

Instructor*Northwestern University**Spring 2024*

Computing Everywhere. Human-Computer Interfaces for Musicking (with Annie Chu)

Instructor*Northwestern University**Winter 2024*

Computing Everywhere Workshop. Generative AI (with Julia Barnett)

Teaching Assistant*Northwestern University**Spring 2022*

COMP_SCI 497 – Digital Musical Instrument Design

Teaching Assistant*Northwestern University**Fall 2021*

EECS 349 – Intro to Machine Learning

Teaching Assistant*Georgia Southern University*

2018 - 2019

Electric Circuit Analysis

SERVICE

Reviewer*NeurIPS Creative AI Workshop*

2025

Reviewer*Interspeech 2025*

2025

Reviewer*ISMIR 2024*

2024

Reviewer*CHI 2023*

2023

Reviewer*ICASSP 2022-2025*

2025

Board Member*Latin@CS - Northwestern University**Fall 2021*